



Helping Robots Focus And Lend A Hand

MIT (USA) has developed a system called 'Relevance' to help robots focus on what matters in the real world. By using a camera and microphone, the robot observes its



Using a novel relevance framework developed at MIT, the robot identifies and prioritizes objects in the scene to autonomously assist humans in a seamless, intelligent, and safe manner (Credit: Courtesy of the researchers)

surroundings to predict a person's goals, then quickly offers assistance. In a test at a breakfast buffet, the robot accurately identified goals 90% of the time and selected the right items 96% of the time, reducing accidents by

60%. The system works like the brain's RAS, filtering out unnecessary information and focusing on what's important to assist people effectively.

Cold Sintering Solves Battery Challenge

Penn State University (Pennsylvania) researchers are advancing solid-state batteries to replace traditional lithium-ion versions. They focus on safer, thermally stable solid-state electrolytes (SSEs), long hindered by high processing temperatures and low conductivity. Using cold sintering—a low-temperature, high-pressure method—they have developed a LATP-PILG ceramic-polymer composite. This combines ceramic stability with the high conductivity of a poly-ionic liquid gel. By engineering grain



Penn State researchers recently published their work investigating the application of an advanced manufacturing technique in battery creation (Credit: Hongtao Sun and techxplore.com)

boundaries, they enhanced ion transport and efficiency. The result is a safer, high-conductivity battery with broader voltage capacity. Sun aims to scale this method for wider use in the energy and semiconductor industries.

Nanowire Semiconductors Harvest Sunlight Hydrogen

A University of Alberta (Edmonton, Canada) team, in collaboration with the Technical University of Munich, has developed a method to produce hydrogen fuel using only sunlight and earth-abundant materials. Led by Professor Karthik Shankar, the team used carbon nitride and titanium dioxide to form a semiconductor heterojunction that directly splits water into hydrogen and oxygen. This efficient, photocatalytic process absorbs diffused sunlight, even on cloudy days, and bypasses traditional solar inefficiencies. The materials are inexpensive and scalable, offering a durable, low-cost, and potentially commercial solution for clean, portable hydrogen energy within a few years.

Chip Checks Cooling For Stacked Electronics

As microelectronic systems become more powerful, the industry is stacking chips for 3D integration to place pro-



This silicon wafer contains chips designed to test cooling systems for 3D integrated microelectronics. Each chip comprises circuitry that generates heat within a 3D stack and measures temperature as cooling solutions are applied (Credit: Nicole Fandel)

cessors next to specialised chips for tasks like AI, communication, and imaging. However, this leads to heating problems. To address this, MIT (USA) Lincoln Laboratory developed a chip that simulates heat generation and measures temperature changes to test cooling solutions. The chip mimics both background heat

and localised hot spots in 3D chip stacks, helping researchers understand how heat spreads and evaluate cooling methods like microchannels delivering cold liquid.

Passivation Boosts Kesterite Solar Efficiency

Researchers from Shenzhen University (China) and the University of Rennes (France) have developed a heat-treatment method for the CdS/CZTS interface in an oxygen-rich environment. Kesterite-based compounds like $\text{Cu}_2\text{ZnSnS}_4$ (CZTS) are eco-friendly and affordable but suffer from low power conversion efficiencies due to atomic-scale defects. This passivates sulphur vacancies and enhances performance by forming beneficial complexes. Their technique achieved a certified 11.51% efficiency, advancing kesterite photovoltaics as a greener, more competitive alternative to silicon-based solar cells.